



Invited Review Article

Exercise lactate in post-COVID-19 condition: Pathophysiological signal, phenotyping tool, or candidate biomarker? A narrative review with a hypothesis-generating clinical observation

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ABSTRACT

Post-COVID-19 condition (PCC), commonly termed long COVID, is a heterogeneous multisystem disorder in which fatigue, exertional dyspnoea, post-exertional symptom exacerbation, and reduced exercise tolerance are prominent. No single laboratory measurement currently confirms or excludes PCC. Because lactate integrates glycolytic flux, pyruvate oxidation, muscle recruitment, oxygen delivery and extraction, adrenergic drive, and clearance by the liver and other tissues, exercise-associated lactate has attracted interest as a potential marker of impaired bioenergetics in PCC. This narrative review evaluates the physiological rationale and clinical evidence for lactate assessment at rest and during exercise in adults with PCC and places a descriptive observation from 22 patients assessed in our clinic into that context. Cardiopulmonary exercise testing studies demonstrate reduced peak oxygen uptake in many symptomatic individuals, although reported mechanisms include deconditioning, dysfunctional breathing, chronotropic incompetence, preload failure, impaired systemic oxygen extraction, autonomic dysfunction, and peripheral or mitochondrial abnormalities. Small mechanistic studies have reported higher exercise lactate, reduced calculated fat oxidation, altered skeletal-muscle metabolism, or more importantly impaired mitochondrial function. In our uncontrolled clinical series, mean arterial lactate increased from 1.23 ± 0.40 mmol/L at rest to 7.11 ± 2.98 mmol/L at peak exercise; the marked difference between testing protocols underscores the methodological dependence of peak values. Overall findings are heterogeneous, populations are selected, and lactate is strongly dependent on achieved work rate, exercise duration, phenotype, medications, nutritional state, and sampling time. Peak lactate alone therefore lacks the specificity, standardization, and validated thresholds required for diagnosis. Its most promising role is as one component of a standardized metabolic exercise phenotype, interpreted alongside work rate, oxygen uptake, ventilatory thresholds, respiratory exchange ratio, symptoms, and recovery kinetics. Controlled prospective studies are required before clinical implementation.

Introduction

Post-COVID-19 condition (PCC) is defined by the World Health Organization as symptoms occurring in individuals with a history of

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